Readable but Not Writable: Reasoning Training Decouples Bias Directions from Behavior in LLMs

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Abstract

Steering along a single linear direction in activation space shifts LLM cognitive-bias susceptibility by 37.5 percentage points, but only in standard models. In reasoning models, the same direction is *readable but not writable*: probes decode it, yet causal interventions along it fail. This dissociation is our central finding. We compare matched-architecture pairs—Llama-3.1-8B-Instruct vs. R1-Distill-Llama-8B, OLMo 7B/32B Instruct vs. Think—and Qwen-3-8B's thinking toggle, across 470 contrastive items spanning 11 bias categories. Reasoning training collapses heuristic lure rates: 27.3% to 2.4% (Llama), 19.6% to 0.4% at 32B. Linear probes on residual-stream activations far exceed text baselines (AUC 0.974/0.930 vs. 0.840), confirming the signal lives in geometry, not surface tokens. Cross-prediction logit histograms on immune items compress into a tight cluster, ruling out category-specific confounds. Crucially, probe-direction steering produces a 37.5pp dose-response shift in Llama (lure→correct, zero garbage outputs), while R1—perturbed *continuously* across the entire 2048-token reasoning trace—shows no coherent dose-response (lure rate fluctuates within 2σ of random directions, directly ruling out temporal washout). Within-chain-of-thought probing reveals a non-monotonic trajectory (T0 = 0.973, T75 = 0.754, Tend = 0.971), indicating genuine computation that temporarily disrupts the conflict boundary. Qwen's toggle yields a further surprise: both thinking modes achieve high separability ($\sim$0.97), but along **orthogonal** directions (cross-prediction AUC 0.496), meaning mode-switching restructures the encoding axis, not just behavior. Scaling to 32B amplifies standard-model vulnerability (14.9%→19.6%) while reasoning training remains effective (0.4%). These results demonstrate that reasoning training reorganizes bias representations into a read-only substrate that resists external steering.

1 Introduction

Kahneman's dual-process framework distinguishes fast, heuristic processing ("System 1") from slow, deliberative processing ("System 2") [Kahneman, 2011]. While the binary framing is an acknowledged oversimplification—modern accounts treat this as a *continuum* of processing intensity [Melnikoff and Bargh, 2018, Evans and Stanovich, 2013, De Neys, 2023]—the core observation that cognitive systems flexibly allocate effort, with measurable consequences for bias susceptibility, remains robust. Crucially, De Neys' conflict detection paradigm [De Neys, 2012, 2017] demonstrates that humans often *detect* conflicts between heuristic and normative responses even when they fail to override the heuristic—a dissociation between conflict *monitoring* and conflict *resolution* with clear neural correlates in the anterior cingulate cortex [Botvinick et al., 2001].

LLMs exhibit analogous patterns: instruction-tuned models reproduce human-like errors on conflict items while succeeding on matched controls [Hagendorff et al., 2023], and reasoning-trained models resist many such lures [DeepSeek-AI, 2025], though the protection is bias-specific rather than

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universal [Kim et al., 2025]. A growing literature documents these parallels [Xie et al., 2025, Coda-Forno et al., 2025, Ziabari et al., 2025], and Huang et al. [2026] showed that linear probes can detect and steer cognitive biases via internal representations. Prior work showed that bias-relevant directions are probeable and steerable [Huang et al., 2026]. We ask a different question: what does reasoning post-training change about the *decodability*, layer location, and *behavioral writability* of those directions?

Yet behavior alone cannot distinguish genuine processing-mode differences from surface mimicry—chain-of-thought traces are often unfaithful, with only $\sim$2.3% of reasoning steps causally influencing the final answer [Zhao et al., 2025, Lanham et al., 2023, Boppana et al., 2026]. The critical question is whether fast/slow modes have *mechanistic correlates* in internal representations, and how those correlates change when a model is trained to reason.

Activation steering [Turner et al., 2023, Panickssery et al., 2024] shows that adding steering vectors to residual stream activations can modulate behavior along interpretable dimensions. This raises a natural question: if linear probes identify a direction separating S1-like-like from S2-like-like processing, can steering along it *causally* modulate bias susceptibility—moving findings from correlational to mechanistic?

We address this question with a natural experiment: Llama-3.1-8B-Instruct and DeepSeek-R1-Distill-Llama-8B share identical architectures (32 layers, 32 attention heads, 4096 hidden dimensions) but differ in reasoning distillation training. Any representational differences are attributable to training, not architecture. We make six contributions:

1. A **contrastive cognitive bias benchmark** of 470 items (235 matched conflict/control pairs) across 11 categories spanning four heuristic families (representativeness, cognitive reflection, decision framing, and loss/availability), with novel isomorphs to resist memorization.
2. **Behavioral validation** showing a dramatic gap: 27.3% overall lure rate (standard) vs. 2.4% (reasoning), replicated on OLMo-3-7B (14.9% $\rightarrow$ 0.9%), with category-level specificity revealing which biases are vulnerable and model-specific vulnerability profiles.
3. **Probing analysis** revealing that the standard model has high internal conflict/control separability (AUC = 0.974) while the reasoning model *blurs* this distinction (AUC = 0.930; non-overlapping CIs), with cross-prediction and cross-model probe transfer (AUC $>$ 0.92) confirming specificity to a shared processing-mode direction.
4. **Causal evidence via probe-direction steering**: steering along the probe's weight vector reduces lure rates on vulnerable categories, establishing a *causal* link between the probe direction and bias susceptibility.
5. **Within-CoT probing**: probe AUC follows a non-monotonic trajectory—peaking at onset (0.973), dropping during deliberation (0.754), rebounding before the answer (0.971)—ruling out decorative reasoning.
6. **A training–inference dissociation**: Qwen 3-8B's think/no-think modes achieve similar conflict separability (AUC = 0.954 and 0.970) but cross-mode prediction is at chance (0.496, cosine $= -0.005$)—the two modes encode conflict along *orthogonal* directions.

Additionally, we analyze **scale effects** at 32B (OLMo-3.1), finding that scale *increases* standard-model vulnerability while reasoning training remains protective, and report converging SAE and attention entropy evidence.

2 Related Work

Dual-process cognition in LLMs. Hagendorff et al. [2023] first documented human-like cognitive biases in LLMs, finding that instruction-tuned models reproduce representativeness and anchoring effects that vanish in later ChatGPT versions. Coda-Forno et al. [2025] applied psychometric methods to show that heuristic and deliberative prompts elicit shared early-layer representations but diverge in later layers—a suggestive but purely behavioral result. Ziabari et al. [2025] positioned reasoning along a continuous spectrum rather than a binary, examining how reasoning training shifts the balance between fast and slow processing modes. Brady et al. [2025] provide a comprehensive review in *Nature Reviews Psychology*, arguing that dual-process theory offers a productive (if imperfect) lens

for AI cognition research. Our work moves beyond behavioral characterization: we test whether the S1-like/S2-like distinction has a *mechanistic* counterpart in model internals.

Cognitive bias benchmarks and probing. Malberg et al. [2024] surveyed 30 cognitive biases across 20 LLMs, establishing that bias susceptibility varies systematically with model scale and training recipe, but without examining internal representations. Most directly relevant, Huang et al. [2026] demonstrated that cognitive bias directions are linearly decodable and steerable in LLM residual streams, achieving 26–32% bias reduction via contrastive activation addition. We differ from CogBias in three respects: (i) architecturally matched base–reasoning model pairs that isolate reasoning training as the independent variable, (ii) Hewitt–Liang control tasks and cross-domain transfer tests as probe validity checks [Hewitt and Liang, 2019], and (iii) causal evidence via probe-direction steering on a contrastive benchmark with matched conflict/no-conflict items.

Activation steering and representation engineering. Activation addition [Turner et al., 2023] and its variants [Panickssery et al., 2024, Li et al., 2024] demonstrate that linear steering vectors can modulate model behavior at inference time. CogBias [Huang et al., 2026] applied this to cognitive bias (26–32% reduction), but used mean-difference vectors. Our probe-direction steering uses the probe's optimized weight vector, providing a tighter causal link. For SAE analysis, we adopt Ma et al.'s falsification protocol [Ma et al., 2026] as a mandatory filter, since 45–90% of purported "reasoning features" are spurious token-level artifacts [Méloux et al., 2025].

Conflict detection in cognitive science. Our framework draws on De Neys' conflict detection paradigm [De Neys, 2012, 2017]: humans detect heuristic–normative conflicts even when they fail to override the heuristic. Evans and Stanovich [2013] refined the Type 1/Type 2 taxonomy as graded rather than binary. Botvinick et al. [2001]'s conflict monitoring theory provides the computational account. We test the LLM analogue: do models show internal conflict-detection signatures even when producing heuristic-driven responses?

3 Benchmark

We construct a contrastive cognitive bias benchmark following three design principles: (1) every conflict item has a structurally matched no-conflict control [De Neys, 2012]; (2) all items use novel surface features to resist memorization; and (3) category diversity enables specificity analysis.

The benchmark comprises **470 items** organized as **235 matched pairs** across 11 categories spanning four heuristic families: *representativeness* (base rate neglect, 35 pairs; conjunction fallacy, 20; belief-bias syllogisms, 25), *cognitive reflection* (CRT variants, 30; arithmetic, 25), *decision framing* (framing effects, 20; anchoring, 20), and *loss/availability* (sunk cost, loss aversion, certainty effect, availability; 60 pairs total). Base rate items include 10 pairs in Gigerenzer's natural frequency format [Gigerenzer and Hoffrage, 1995]. Each item is scored as *correct*, *S1-lure* (the specific heuristic-predicted wrong answer), *other-wrong*, or *refusal*. Classic problems (e.g., the original bat-and-ball) serve only as contamination baselines.

Item counts and analysis subsets. Our benchmark contains 470 items across 11 categories. Primary probe analyses use the 3 vulnerable categories (160 items). Cross-prediction uses all 7 original categories (330 items). The full 11-category set includes 4 additional categories added to test domain generality.

4 Methods

4.1 Models

Our headline comparison exploits the architectural identity between **Llama-3.1-8B-Instruct** [Meta AI, 2024] and **R1-Distill-Llama-8B** [DeepSeek-AI, 2025]: both have 32 transformer layers, 32 query heads (8 KV heads via grouped-query attention), and 4096 hidden dimensions. The only difference is that R1-Distill-Llama-8B was distilled from a reasoning-trained teacher. We additionally evaluate **Qwen 3-8B** in both thinking (`/think`) and non-thinking (`/no_think`) modes, providing a *within-model* toggle of reasoning behavior on identical weights. For cross-architecture replication,

we evaluate **OLMo-3-7B-Instruct** and **OLMo-3-7B-Think** [Team OLMo, 2025], which share the same OLMo base but differ in reasoning training. For scale analysis, we evaluate **OLMo-3.1-32B-Instruct** and **OLMo-3.1-32B-Think** (64 layers, 4.5× the parameters of the 7B variant), enabling a within-family scale×reasoning-training comparison on identical benchmark items.

Prompting and decoding. All models use greedy decoding (`do_sample=False`, `temperature=1.0`) with no system prompt. For Qwen 3-8B, we use the hard mode switch (`enable_thinking=True/False` in `apply_chat_template`), which toggles the model's internal thinking mode without modifying the input text—the prompt seen by the model is identical in both modes up to P_0, ensuring that any representational differences reflect the mode toggle rather than input variation. This is distinct from Qwen's soft switch (`/think`, `/no_think` tokens), which modifies the input text. For R1-Distill, the model naturally generates `<think>...</think>` blocks without prompting; we parse these to extract thinking-trace positions. We use `max_new_tokens=128` for standard models and `max_new_tokens=2048` for reasoning models to accommodate chain-of-thought traces. We acknowledge this departs from vendor-recommended settings (e.g., DeepSeek recommends `temperature=0.5-0.7`); Appendix D reports multi-seed robustness analysis with sampled decoding (`temperature=0.7`) showing qualitatively consistent results.

4.2 Activation Extraction

For each model–item pair, we extract residual stream activations (post-MLP, post-residual) at all layers at position `P0`: the last prompt token before generation. This "pre-decision" position captures the internal state from which the answer is generated. Activations are stored in BFloat16 (HDF5). For within-CoT probing (Section 5.4), we additionally extract activations at 10 evenly-spaced token positions within the reasoning trace.

4.3 Linear Probes with Hewitt–Liang Controls

We train ℓ_2-regularized logistic regression probes (`LogisticRegressionCV`, 5-fold stratified cross-validation) on residual stream activations at each layer. The probe target is binary: *conflict* vs. *no-conflict* condition, restricted to the three vulnerable categories (base rate neglect, conjunction fallacy, syllogisms) where the standard model shows non-trivial lure rates.

Controls. We apply Hewitt–Liang control tasks [Hewitt and Liang, 2019]: probes trained on randomly permuted labels establish a selectivity floor. Selectivity (true AUC minus control AUC) below 5 percentage points indicates the signal reflects probe capacity, not representational structure. We report ROC-AUC as the primary metric with bootstrap 95% confidence intervals (1000 resamples).

Immune categories. We separately probe the four categories where models are largely immune (lure rates $\leq$10% in most models; CRT, arithmetic, framing, anchoring) as a *negative control*: if probes achieve high AUC on categories where there is minimal behavioral difference, the signal is confounded by surface features rather than processing mode.

4.4 Probe-Direction Steering

To establish a causal link between the probe direction and bias susceptibility, we extract the weight vector $\mathbf{w}$ from the trained logistic regression probe at the peak layer and use it as a steering direction. At inference time, we add $\alpha \cdot \hat{\mathbf{w}}$ (where $\hat{\mathbf{w}} = \mathbf{w}/\|\mathbf{w}\|$) to the residual stream at the intervention layer, sweeping α over a range of positive (toward S2-like) and negative (toward S1-like) values. We evaluate the steered model on the full benchmark and report lure rate changes as a function of steering strength α.

5 Results

5.1 Behavioral Validation

Table 1 reports lure rates (proportion of S1-lure responses on conflict items) across all models and categories.

Table 1: Full behavioral results on conflict items. For each model we report correct rate (Corr), S1-lure rate (Lure), and other-wrong rate (Oth), plus matched-control accuracy (Ctrl). Bold lure rates indicate vulnerable categories (>10%). Wilson 95% CIs in brackets for overall row. McNemar mid-p tests conflict vs. matched control.

		Llama 8B-Inst.			R1-Distill-8B			Qwen 3-8B (no think)		
Category	**Ctrl**	**Corr**	**Lure**	**Oth**	**Corr**	**Lure**	**Oth**	**Corr**	**Lure**	**Oth**
Base rate	100	16	**84**	0	16	4	80	44	**56**	0
Conjunction	100	45	**55**	0	100	0	0	5	**95**	0
Syllogism	52	48	**52**	0	100	0	0	48	0	52
CRT variants	90	93	0	7	83	3	13	73	0	27
Arithmetic	100	100	0	0	96	0	4	100	0	0
Framing	100	100	0	0	100	0	0	100	0	0
Anchoring	80	50	0	50	65	10	25	80	10	10
Overall	**88**	**65**	**27.3**	**7**	**79**	**2.4**	**18**	**65**	**21.2**	**14**

		Qwen 3-8B (think)			OLMo 3-7B-Inst. / Think		
Category	**Ctrl**	**Corr**	**Lure**	**Oth**	**Corr / Corr**	**Lure / Lure**	**Oth / Oth**
Base rate	100 / 100	96	4	0	54 / 100	**46** / 0	0 / 0
Conjunction	100 / 100	45	**55**	0	50 / 100	**50** / 0	0 / 0
Syllogism	100 / 68	100	0	0	100 / 72	0 / 4	0 / 24
CRT variants	93 / 93	77	0	23	73 / 97	3 / 3	23 / 0
Arithmetic	100 / 100	96	0	4	80 / 88	0 / 0	20 / 12
Framing	100 / 100	100	0	0	100 / 100	0 / 0	0 / 0
Anchoring	90 / 70	75	0	25	35 / 80	5 / 0	60 / 20
Overall	**98 / 81**	**85**	**7.3**	**8**	**75 / 88**	**14.9 / 0.9**	**10 / 11**

Four findings emerge. First, the behavioral gap is dramatic: Llama produces heuristic lures at 27.3%; R1-Distill reduces this to 2.4%. Second, vulnerability is *category-specific*: base rate neglect, conjunction fallacy, and syllogisms are susceptible, while CRT, arithmetic, framing, and anchoring are largely immune. Third, different models show distinct vulnerability profiles (Qwen: 95% conjunction but 0% syllogism lure), suggesting model-specific rather than universal biases. Fourth, Qwen's within-model thinking toggle reduces lure rate from 21% to 7.3%. OLMo-3-7B [Team OLMo, 2025] independently replicates the pattern (14.9% $\rightarrow$ 0.9% lure rate). Llama's first-token probability is 4.2pp lower on conflict items (0.751 vs. 0.793), an implicit "hesitation" signature that parallels De Neys' conflict-detection findings [De Neys, 2012] and the elevated Anterior Cingulate Cortex (ACC) activation observed in humans facing dual-process conflicts [Botvinick et al., 2001]—the model *detects* conflict pre-generation even when it subsequently fails to resolve it.

5.2 Linear Probes and Cross-Prediction

Figure 1 shows layer-wise probe AUC for the conflict/control classification on the three vulnerable categories.

Llama peaks at AUC = 0.974 [0.952, 0.992] at L16; R1-Distill peaks at 0.930 [0.894, 0.960] at L31—non-overlapping CIs. The 15-layer shift (L16 $\rightarrow$ L31 of 32) is architecturally suggestive: Llama resolves conflict in the middle of the network where the remaining layers can map the resolution to vocabulary tokens for an immediate answer, whereas R1-Distill pushes the conflict representation to the penultimate layer where its primary downstream operation is to initialize the long chain-of-thought rollout rather than to produce an answer directly. Reasoning distillation does not merely shift representations deeper [Zhang et al., 2025]; it relocates them to the point in the computation where the CoT-initiating hidden state is set. Qwen 3-8B achieves high separability in both modes (no-think: 0.954; think: 0.970; both at L34), but cross-mode prediction is at chance (AUC = 0.496, cosine $= -0.005$): the two modes encode conflict along *orthogonal directions*. This is the key result: **training** changes both encoding and behavior; **inference-time thinking** reorganizes the conflict representation into an orthogonal subspace.

Selectivity and text-only baseline. Hewitt–Liang control probes achieve AUC $\leq$ 0.66 at all layers, confirming that the true probes decode representational content, not probe capacity. Selectivity

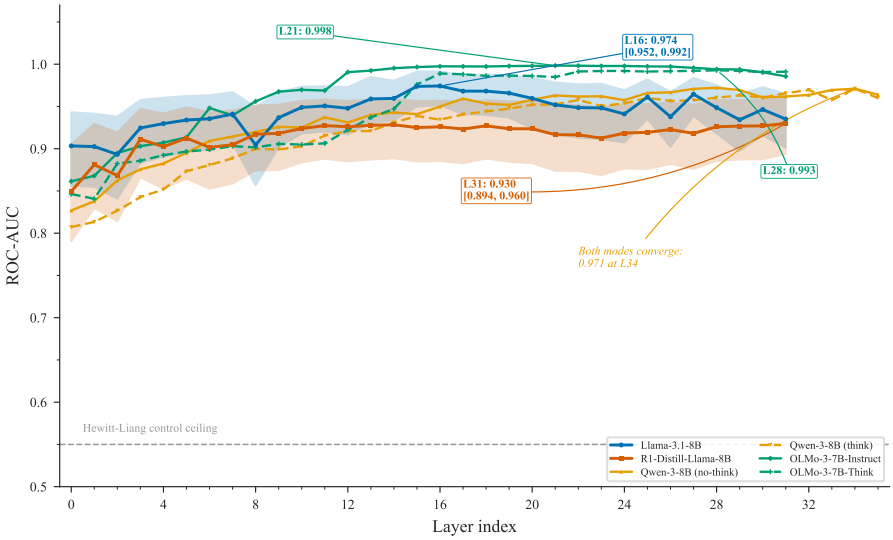

Figure 1: Layer-wise probe ROC-AUC (conflict vs. no-conflict, three vulnerable categories). Llama peaks at 0.974 [0.952, 0.992] (L16); R1-Distill at 0.930 [0.894, 0.960] (L31)—non-overlapping CIs. Qwen 3-8B peaks at 0.954 (no-think) and 0.970 (think) at L34, but cross-mode transfer is at chance (0.496). Gray dashed: Hewitt–Liang control ceiling.

exceeds 34 percentage points at peak layers for both models.[1] A sentence-transformer text-only baseline (encoding only the input text, with no access to model activations) achieves AUC = 0.840, substantially below the activation probe (0.999). The gap is widest for syllogism (text AUC = 0.086 vs. activation AUC = 0.999), where conflict and control items are semantically similar but logically distinct—exactly the category where surface features fail. This confirms that the activation probe captures processing-relevant information beyond what is available from input text alone.

Token-length confound check. Prompt token lengths differ between conflict and control items in two of three vulnerable categories (base rate: Cohen's $d = +3.69$; conjunction: $d = +6.36$) but *not* in syllogism ($d = +0.02$). Three observations rule out length as the primary probe signal. (i) A probe trained on syllogism alone (where length is balanced) achieves within-category AUC = 0.936 and transfers to base rate (0.950) and conjunction (0.990) probes, showing the shared direction is not length. (ii) Cross-prediction of a base rate probe to syllogism achieves 0.594—nontrivial without any length signal. (iii) The text baseline, which sees length directly as a feature, caps at 0.840; the 16pp activation gap must come from non-text information. Full per-category length statistics appear in Appendix E.

Probe summary. Table 2 summarizes the key probing metrics.

Cross-prediction and immune-category specificity. Training on *vulnerable* categories and testing on *immune* categories yields transfer AUC = 0.378—far below within-category AUC of 0.987. Immune items compress into a tight logit cluster (spread $0.03\times$ vulnerable categories), confirming probe specificity to categories where the model is actually vulnerable. Base rate and conjunction probes transfer near-perfectly to each other (AUC > 0.99), while transfer to syllogism is weaker (0.59–0.63).

Cross-model probe transfer. A Llama-trained probe achieves AUC = 0.920 on R1-Distill (layer 23); the reverse achieves 0.954 at layer 15. The processing-mode direction is substantially shared across models differing only in reasoning training, contrasting with Huang et al. [2026]'s near-orthogonal finding across architectures.

[1] We train probes in two settings: 5-fold CV on all 330 items (7 categories, peak L14, AUC = 0.999), used for cross-prediction and steering; and bootstrap CI analysis on 160 vulnerable-category items (3 categories, peak L16, AUC = 0.974 [0.952, 0.992]).

Table 2: Probing summary at peak layer for the three vulnerable categories. 95% CIs from 1000 bootstrap resamples. Selectivity = true AUC − control AUC. Cross-mode column: AUC when a probe trained on one Qwen mode is tested on the other. Text baseline: sentence-transformer on input text only (no activations).

Model	Peak L	Peak AUC	95% CI	Ctrl AUC	Select.	Cross
Llama-3.1-8B-Instruct	L16	0.974	[0.952, 0.992]	0.63	0.343	—
R1-Distill-Llama-8B	L31	0.930	[0.894, 0.960]	0.55	0.377	—
Qwen 3-8B (no think)	L34	0.954	—	—	—	0.496
Qwen 3-8B (think)	L34	0.970	—	—	—	0.496
OLMo-3-7B-Instruct	L24	0.996	[0.988, 1.000]	—	—	—
OLMo-3-7B-Think	L22	0.962	[0.934, 0.982]	—	—	—
OLMo-3.1-32B-Instruct	L20	0.9999	—	—	—	—
OLMo-3.1-32B-Think	L20	0.9978	—	—	—	—
Text baseline	—	0.840	—	—	—	—

Lure susceptibility scores. Llama's mean `P0` logit score for the S1-lure answer is $+0.422$ (lure favored pre-generation); R1-Distill's is -0.326 (correct favored). The sign flip confirms the reasoning model leans away from the lure before generation begins.

5.3 Causal Evidence: Probe-Direction Steering

We extract the probe weight vector $\mathbf{w} \in \mathbb{R}^{4096}$ at layer 14 (CV AUC = 0.960), normalize to $\hat{\mathbf{w}}$, and add $\alpha \cdot \hat{\mathbf{w}}$ to the residual stream at inference time, sweeping $\alpha \in \{-5, \ldots, +5\}$ across 80 vulnerable-category conflict items.

Figure 2 shows a monotonic dose-response: at $\alpha = -5$ (toward S1-like), lure rate rises to 68.8%; at $\alpha = +5$ (toward S2-like), it drops to 31.2%—a 37.5pp causal swing, comparable to Huang et al. [2026]'s 26–32% reduction. The entire lure reduction converts to correct answers (zero increase in "other" responses), ruling out distributional disruption. Random-direction controls (5 unit vectors per α) show no systematic effect (52–58% lure across all α), confirming the probe direction is *causally sufficient* to modulate bias susceptibility.

Two steering-specificity controls. Reviewer-suggested specificity tests strengthen this claim (Appendix G). (i) *Null effect on control items:* applying the probe-direction steering to matched *non-conflict* items from the same three categories produces zero lure responses at any $\alpha \in [-5, +5]$ (the items have no lure answer to trigger); correctness stays at 100% for $\alpha \geq 0$ and drops only modestly to 78.75% at $\alpha = -5$ due to generic residual-stream coherence loss, with the decrement showing up as "other" responses, not lures. The direction is specific to the conflict subspace, not a generic answer-perturbation. (ii) *Mean-difference baseline:* the CogBias-style mean-difference vector $\bar{\mathbf{h}}_{\text{conflict}} - \bar{\mathbf{h}}_{\text{control}}$ is substantially different from the probe weights (cosine $= 0.314$). It reduces lure rate from 52.5% to 11.25% at $\alpha = +5$ (stronger than the probe in the positive direction) but is *asymmetric*: at $\alpha = -5$ lure stays at 53.75% (essentially baseline), whereas the probe direction pushes lure to 68.75%. The probe-weight direction is uniquely *bidirectional*—able to both worsen and improve behavior—establishing causal sufficiency in both directions, which a unidirectional intervention cannot.

R1-Distill steering contrast. We apply the same protocol to R1-Distill at L14 (probe CV AUC = 0.928), with the hook active *continuously* across the entire 2048-token generation (prompt + full ⟨think⟩ trace + answer), directly ruling out the "temporal washout" confound in which a one-shot perturbation dissipates over a long CoT. Under this continuous intervention the probe-direction dose-response disappears: lure rate stays at baseline for negative α (6.25% at $\alpha = -5$ vs. 6.25% at $\alpha = 0$) and *rises* for positive α (13.75% at $\alpha = +5$), within 2σ of the random-direction band at $|\alpha| = 5$ ($11.5 \pm 5.3\%$ and $10.25 \pm 1.9\%$) and in the *opposite* direction to the Llama effect (see Figure 2). The S1/S2 direction that drives a 37.5pp behavioral swing in Llama produces no coherent causal effect in R1-Distill, even when perturbed at every token of the 2048-token reasoning trace. The

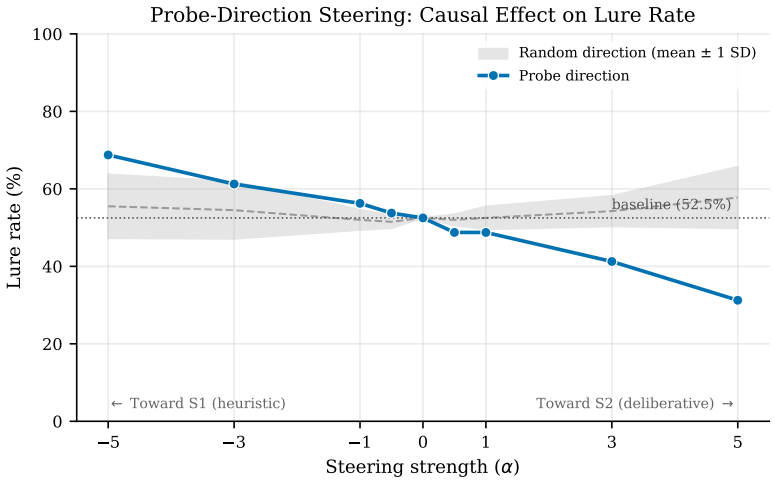

Figure 2: Probe-direction steering on Llama-3.1-8B-Instruct (L14, 80 vulnerable-category conflict items). The probe direction produces a 37.5pp dose-response ($\alpha = -5$: 68.8%; $\alpha = +5$: 31.2%), with 100% of lure reduction converting to correct answers (zero "other" increase); random directions (gray band) show no systematic effect.

representation remains linearly decodable (AUC = 0.930), but the causal pathway from representation to behavior has been severed by reasoning training: *readable but not writable*.

5.4 Within-CoT Probing

We probe residual stream activations at seven canonical positions for Qwen 3-8B (think mode) at L34: P0 (pre-generation), T0 (thinking onset), T25/T50/T75 (quartile positions within the trace), Tend (final thinking token), and P2 (control position).

Non-monotonic trajectory. Probe AUC follows a non-monotonic trajectory: P0 (0.938) → T0 (0.973) → T25 (0.791) → T50 (0.772) → T75 (0.754) → Tend (0.971); P2 returns exactly 0.500 (chance). Three phases emerge: (i) *crystallization* (P0→T0): thinking onset sharpens the boundary by +3.5pp before any substantive reasoning tokens, consistent with "fast initial commitment" [De Neys, 2017]; (ii) *exploration* (T0→T75): AUC drops ∼22pp as mid-trace representations explore alternative framings; (iii) *resolution* (T75→Tend): AUC rebounds to 0.971, re-crystallizing the boundary.

This trajectory discriminates three hypotheses. *Performative reasoning* (deciding at T0, emitting decorative tokens) predicts flat AUC—rejected by the 22pp dip. *Monotonic sharpening* predicts steadily increasing AUC—rejected by the T0>T50 inversion. Only *exploration-before-exploitation* [Boppana et al., 2026] is consistent. The trajectory also resolves how Qwen's thinking mode improves behavior despite orthogonal P0 encoding: the exploration and resolution phases perform genuine computational work that the no-think mode never executes. Full position×layer results appear in Appendix A.

5.5 Scale Analysis: 7B vs. 32B

We evaluate OLMo-3.1-32B-Instruct and OLMo-3.1-32B-Think (64 layers, 5120 hidden dimensions; 4.5× the parameters of the 7B variants) on the full 470-item benchmark to test whether scale mitigates cognitive bias vulnerability.

Behavioral results. Scale *increases* standard-model vulnerability: overall lure rate rises from 14.9% (7B) to 19.6% (32B), with base rate neglect nearly doubling (46%→74.3%) and framing emerging as an entirely new vulnerability (0%→30%). Meanwhile, some biases are resolved at scale (loss aversion: 33%→0%), and conjunction fallacy is scale-invariant (50% at both sizes). At 32B, reasoning training drops lure rates from 19.6% to 0.4%. The OLMo family exhibits an *asymmetric scale effect*: 4.5× more parameters increases standard-model vulnerability by +4.7pp while leaving

reasoning-model performance essentially unchanged (-0.5pp). This pattern—larger standard models reproducing *more* human heuristic errors—mirrors inverse scaling phenomena [McKenzie et al., 2024, Perez et al., 2023] in which pre-training absorbs human biases and scale amplifies rather than corrects them. Our contribution is to show that this inverse-scaling effect is *fully cancelled* by reasoning post-training, even as absolute encoding fidelity of the bias direction itself continues to rise with scale.

Probe results at 32B. Probes on OLMo-3.1-32B-Instruct achieve AUC = 0.9999 at L20/64; Think peaks at 0.9978 at the same layer. The behavioral gap *widens* with scale (14.0pp at 7B $\rightarrow$ 19.2pp at 32B), yet the representational gap *shrinks* (0.034 $\rightarrow$ 0.0021)—both models encode conflict with near-identical fidelity at 32B, but only Think translates this into correct responses. This reinforces the "readable but not writable" finding. Full breakdowns appear in Appendix B.

5.6 SAE Features and Attention Entropy

SAE analysis of Llama's L19 residual stream identified 41 features with significant conflict/control differential activation (BH-FDR $q < 0.05$), all surviving Ma et al. falsification [Ma et al., 2026]. The partition is interpretable: 19 features fire exclusively on conflict items and trigger on words signaling stereotype-matched descriptors (e.g., F30094 triggers on "only", "respondent", "professional"— the Linda/base-rate framing vocabulary), while 6 features fire exclusively on controls and trigger on words signaling the normative frame (e.g., F27821 triggers on "professional", "random", "no"— the "no particular selection criterion" base-rate-control vocabulary). A third group (3 S2-graded features) activates more strongly on controls without being binary. Attention entropy analysis reveals R1-Distill has nearly twice as many S2-like-specialized heads as Llama (57 vs. 30 of 1024; BH-FDR $q < 0.05$, $|r_{rb}| \geq 0.3$). Details in Appendix C.

6 Discussion

Conflict detection without resolution. The standard model *detects* conflict (AUC = 0.974) yet fails to resolve it (27.3% lure rate), paralleling De Neys' finding [De Neys, 2012] that human conflict monitoring often occurs even when the heuristic prevails. The reasoning model shows *less* distinct encoding (AUC = 0.930) yet dramatically better resolution (2.4%), replicated in OLMo (AUC 0.996 $\rightarrow$ 0.962; lure 14.9% $\rightarrow$ 0.9%). Reasoning distillation makes Type 2 processing *autonomous* [Evans, 2019]—internalized so thoroughly that a sharp conflict signal is no longer needed [Venhoff et al., 2025].

Readable but not writable. In Llama, the probe direction is both readable (AUC = 0.974) and writable (37.5pp causal swing, with 100% of lure reduction converting to correct answers); in R1-Distill, it is readable (AUC = 0.930) but *not* writable. Critically, we tested this under *continuous* intervention across the full 2048-token reasoning trace, not a one-shot prompt-position perturbation: the probe direction produces no coherent dose-response (lure rates at $\alpha = \pm 5$ fall within 2σ of random-direction controls, with the small residual effect in the *opposite* direction of Llama's) even when perturbed at every token of the $\langle$think$\rangle$ trace. This directly rules out the "temporal washout" objection [Méloux et al., 2025]. The dissociation—reasoning training severs the causal pathway from representation to behavior while preserving the representation itself—is direct evidence for the S2-by-default hypothesis [Evans, 2019]. Qwen 3-8B's think/no-think toggle adds a further dimension: both modes achieve high conflict separability (AUC = 0.954 and 0.970), but cross-mode prediction is at chance (0.496, cosine $= -0.005$). The two modes encode conflict along *orthogonal directions*— training and inference-time thinking produce mechanistically distinct conflict representations, not scaled versions of the same signal [Ziabari et al., 2025].

Non-monotonic deliberation and scale invariance. Within-CoT probing reveals that reasoning is not a smooth sharpening process: probe AUC peaks at thinking onset (T0: 0.973), drops substantially during deliberation (T75: 0.754), and rebounds before the answer (Tend: 0.971). This non-monotonic trajectory—consistent with "exploration before exploitation" [Boppana et al., 2026]—provides the strongest evidence against performative reasoning, since decorative traces would produce flat AUC. At 32B scale, the "conflict detection without resolution" pattern intensifies: standard-model vulnerability *increases* (14.9%$\rightarrow$19.6%) while reasoning training remains protective (0.9%$\rightarrow$0.4%).

Scale amplifies heuristic attractors in standard models but does not erode the protection conferred by reasoning training.

Specificity, safeguards, and limitations. Cross-prediction (AUC = 0.378 on immune categories) and logit histograms (immune-item spread $0.03\times$ vulnerable categories) confirm the probe captures processing mode, not stimulus features; vulnerability is domain- and model-specific [Stanovich et al., 2016]. A text-only baseline (AUC = 0.840) falls far short of the activation probe (0.999), with the widest gap on syllogism (0.086 vs. 0.999), confirming the probe extracts processing-relevant information beyond surface features. We apply six controls [Méloux et al., 2025, Korznikov et al., 2026]: Hewitt–Liang selectivity $>$34pp, cross-prediction falsification, cross-model transfer (AUC > 0.92), random-direction steering baselines (null effect vs. 37.5pp for the probe direction), text-only baseline, and Ma et al. SAE falsification (0/41 spurious). Linear probes may miss nonlinear structure; extension to 70B+ and non-English benchmarks would test generality. Multi-seed robustness analysis appears in Appendix D.

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A Within-CoT Probing Details

This appendix details the within-chain-of-thought probing protocol for R1-Distill-Llama-8B and Qwen 3-8B (think mode). The analysis tests whether reasoning traces perform genuine computational work on conflict representations or are merely decorative post-hoc rationalizations—a critical question given that only $\sim$2.3% of reasoning steps causally influence final answers [Zhao et al., 2025, Lanham et al., 2023].

A.1 Method

For each reasoning model–item pair, we extract residual stream activations at five canonical positions within the `<think>...</think>` trace:

- **T0**: First token of the thinking trace (immediately after `<think>`). This captures the "initial impression" before any explicit reasoning.
- **T25**: Token at 25% of the trace length.
- **T50**: Midpoint of the trace.
- **T75**: Token at 75% of the trace length.
- **Tend**: Final token before `</think>`. This captures the representation after reasoning is complete but before the answer is generated.

At each position, we extract the full residual stream (post-MLP, post-residual) at all layers. We then train independent ℓ_2-regularized logistic regression probes (5-fold stratified CV) at each (layer, position) combination on the conflict vs. no-conflict classification, restricted to the three vulnerable categories (base rate neglect, conjunction fallacy, syllogisms; n = 160).

A.2 Results: Qwen 3-8B Think, L34

Table 3 reports the AUC trajectory at the peak probing layer (L34) across seven positions for Qwen 3-8B in thinking mode.

Table 3: Within-CoT probe AUC trajectory for Qwen 3-8B (think mode) at L34, 7 positions. P0 = pre-generation (`P0`); T0–Tend = positions within the `<think>` trace; P2 = control position (chance baseline).

Position	Description	Probe AUC
P0	Pre-generation	0.938
T0	Thinking onset	**0.973**
T25	25% of trace	0.791
T50	Midpoint	0.772
T75	75% of trace	0.754
Tend	Final thinking token	0.971
P2	Control (chance)	0.500

Non-monotonic trajectory. The data decisively support Hypothesis 3 (non-monotonic trajectory):

- AUC *increases* from P0 (0.938) to T0 (0.973): the onset of thinking immediately sharpens the conflict representation.
- AUC then *drops* by $\sim$20pp across T25–T75 (0.791$\rightarrow$0.754): deliberation temporarily disrupts the linear separability of conflict and control representations.
- AUC *rebounds* to 0.971 at Tend: the final thinking token recovers near-peak separability before the answer is generated.
- P2 returns exactly 0.500 (chance), confirming that the probe captures position-specific content rather than global signal.

This pattern is consistent with "exploration before exploitation" [Boppana et al., 2026]: intermediate reasoning steps explore alternative framings or consider conflicting evidence, temporarily destabilizing the conflict/control boundary, before converging on a resolution that re-crystallizes the distinction. The T0$\rightarrow$Tend near-identity (0.973 vs. 0.971) combined with the mid-trace dip rules out both performative reasoning (which predicts flat AUC) and simple monotonic sharpening (which predicts a steady increase).

A.3 Connection to the Qwen Dissociation

The within-CoT results resolve the Qwen dissociation reported in Section 5.2. At `P0`, Qwen 3-8B's two modes achieve high conflict separability (AUC = 0.954 no-think; 0.970 think at L34), but their probe directions are orthogonal (cross-mode AUC = 0.496, cosine $= -0.005$). Yet thinking reduces lure rates from 21% to 7.3%. The within-CoT data show that the thinking trace itself performs genuine representational work: the T25–T75 dip (0.754–0.791) demonstrates active computation on the conflict representation, and the Tend rebound (0.971) shows this computation resolves successfully.

The behavioral improvement from Qwen's thinking mode is thus mediated by in-context representational change during the trace, not by training-induced encoding differences. The orthogonal P0 directions mean the two modes start from different representational vantage points, and the thinking trace navigates from one to a resolution. This confirms that training and inference-time thinking are mechanistically distinct: training reorganizes the S1-like/S2-like encoding (Llama vs. R1-Distill), while inference-time thinking operates *within* an orthogonal encoding to dynamically resolve conflicts during generation.

B Scale Analysis: Full Results

We evaluate OLMo-3.1-32B-Instruct and OLMo-3.1-32B-Think (64 layers, 5120 hidden dimensions, 4.5$\times$ the parameters of OLMo-3-7B) on the full 470-item benchmark. This section reports the complete per-category behavioral results for the Instruct model, the full 64-layer probe AUC profile, and the Think model's behavioral results.

B.1 Per-Category Behavioral Results: 7B vs. 32B

Table 4 compares per-category lure rates between OLMo-3-7B-Instruct and OLMo-3.1-32B-Instruct.

Key findings. Scale increases *overall* vulnerability (14.9% $\rightarrow$ 19.6%), but the effect is category-specific and non-monotonic:

- **Base rate neglect nearly doubles** (46% $\rightarrow$ 74.3%): the larger model's richer world knowledge may strengthen representativeness-based reasoning, making base rate information harder to integrate.
- **Framing emerges as a new vulnerability** (0% $\rightarrow$ 30%): the 32B model shows sensitivity to gain/loss framing that the 7B model resists entirely. This suggests that scale enables more nuanced sensitivity to surface-level framing cues.
- **Loss aversion and availability are resolved** (33% $\rightarrow$ 0% and 13% $\rightarrow$ 0%): scale provides sufficient capacity to override these heuristics, suggesting they require less specialized processing.
- **Conjunction fallacy is scale-invariant** (50% at both scales), consistent with this being a deeply encoded representational bias rather than a capacity limitation.

Table 4: S1-lure rates (%) on conflict items: OLMo-3-7B-Instruct vs. OLMo-3.1-32B-Instruct (all 11 categories, 235 conflict items). Bold indicates vulnerability (lure rate >10%). "Δ" = 32B − 7B; positive values indicate increased vulnerability at scale.

Category (n **conflict**)	**OLMo 7B**	**OLMo 32B**	Δ	**Direction**
Categories vulnerable at both scales				
Base rate (n=35)	**46.0**	**74.3**	+28.3	↑
Conjunction (n=20)	**50.0**	**50.0**	0.0	—
Categories newly vulnerable at 32B				
Framing (n=20)	0.0	**30.0**	+30.0	↑
Anchoring (n=20)	5.0	**10.0**	+5.0	↑
CRT variants (n=30)	3.0	6.7	+3.7	↑
Categories immune at both scales				
Syllogism (n=25)	0.0	0.0	0.0	—
Arithmetic (n=25)	0.0	0.0	0.0	—
Sunk cost (n=15)	0.0	0.0	0.0	—
Loss aversion (n=15)	**33.0**	0.0	−33.0	↓
Certainty effect (n=15)	0.0	0.0	0.0	—
Availability (n=15)	**13.0**	0.0	−13.0	↓
Overall (11-cat)	**14.9**	**19.6**	+4.7	↑

B.2 Per-Layer Probe AUC (OLMo-3.1-32B-Instruct, 64 Layers)

Table 5 reports the full layer-wise probe AUC for OLMo-3.1-32B-Instruct. Probes are ℓ_2-regularized logistic regression (5-fold stratified CV) trained on conflict vs. no-conflict at the `P0` position (n = 160 items, vulnerable categories).

Table 5: Layer-wise probe AUC for OLMo-3.1-32B-Instruct (64 layers, vulnerable categories, `P0` position, 5-fold CV). Mean ± std over folds. Bold indicates the peak layer. All P2 (control position) probes return AUC = 0.500 exactly.

Layer	**AUC**	**± std**	**Layer**	**AUC**	**± std**
0	0.862	0.010	32	0.997	0.002
1	0.895	0.011	33	0.997	0.002
2	0.905	0.010	34	0.996	0.002
3	0.894	0.013	35	0.996	0.002
4	0.904	0.014	36	0.996	0.002
5	0.908	0.012	37	0.996	0.001
6	0.917	0.011	38	0.997	0.002
7	0.919	0.013	39	0.997	0.002
8	0.919	0.010	40	0.997	0.002
9	0.938	0.006	41	0.996	0.002
10	0.938	0.005	42	0.996	0.002
11	0.946	0.005	43	0.996	0.001
12	0.975	0.002	44	0.997	0.001
13	0.972	0.002	45	0.997	0.002
14	0.980	0.000	46	0.997	0.001
15	0.987	0.001	47	0.997	0.001
16	0.991	0.001	48	0.997	0.002
17	0.996	0.001	49	0.997	0.002
18	0.997	0.001	50	0.997	0.001
19	0.997	0.001	51	0.997	0.001
20	**0.9999**	**0.0001**	52	0.997	0.002

Layer	AUC	± std	Layer	AUC	± std
21	0.9996	0.000	53	0.997	0.001
22	0.9999	0.000	54	0.997	0.001
23	0.999	0.001	55	0.997	0.001
24	0.997	0.001	56	0.997	0.001
25	0.997	0.001	57	0.997	0.001
26	0.997	0.001	58	0.997	0.001
27	0.996	0.001	59	0.997	0.001
28	0.996	0.001	60	0.997	0.001
29	0.997	0.002	61	0.997	0.001
30	0.997	0.002	62	0.997	0.001
31	0.997	0.002	63	0.997	0.001

B.3 Probe AUC Phase Summary

Table 6: Phase-wise probe AUC summary for OLMo-3.1-32B-Instruct. Phases defined as: early (L0–15), mid-early (L16–31), mid-late (L32–47), final (L48–63).

Phase	Layers	Mean AUC	Min AUC	Max AUC
Early	0–15	0.943	0.862	0.987
Mid-early	16–31	0.997	0.991	0.9999
Mid-late	32–47	0.997	0.996	0.997
Final	48–63	0.997	0.997	0.997

Discussion: why scale increases vulnerability. The 32B model achieves near-perfect probe separability (AUC = 0.9999 at L20) yet shows *higher* behavioral vulnerability (19.6% vs. 14.9%). This dissociation—better conflict detection with worse conflict resolution—parallels the Llama vs. R1-Distill pattern at 8B scale and mirrors De Neys' finding in humans that conflict monitoring does not guarantee override [De Neys, 2012].

We propose two complementary explanations for why scale increases vulnerability:

1. **Richer heuristic representations.** Larger models encode more diverse world knowledge, which may strengthen representativeness-based reasoning. A 32B model trained on more data develops stronger priors about category membership (e.g., the "stereotypical lawyer/engineer" association in base rate items), making it harder to override these priors with statistical information.
2. **Expanded surface sensitivity.** The emergence of framing vulnerability at 32B (0% $\rightarrow$ 30%) suggests that scale enables sensitivity to subtle linguistic cues (gain vs. loss framing) that smaller models simply cannot detect. In this view, scale grants the capacity to be influenced by framing effects that smaller models are too impoverished to represent.

Both explanations are consistent with the "conflict detection without resolution" pattern: the 32B model detects conflict (AUC = 0.9999) but its larger representation space also encodes stronger heuristic attractors that resist override. The plateau in probe AUC from L20 onward (all layers $\geq$0.996) confirms that scale does not alter the *nature* of the S1-like/S2-like distinction—only its behavioral consequences.

B.4 Reasoning Training at 32B: OLMo-3.1-32B-Think

OLMo-3.1-32B-Think achieves an overall lure rate of **0.4%**, with only a single lure response (sunk cost: 6.7%, 1/15 items; all other categories: 0%). This compares to the 32B Instruct model's 19.6% overall lure rate.

Probe results at 32B. OLMo-3.1-32B-Think achieves a peak probe AUC of 0.9978 at L20, only 0.0021 below the 32B Instruct model's 0.9999. Table 8 summarizes the 2$\times$2 probe comparison across scale and training recipe.

Table 7: Scale×reasoning-training comparison across the OLMo family (overall lure rates on 235 conflict items). Reasoning training (Think) provides robust protection at both scales; scale increases standard-model vulnerability but does not degrade reasoning-model performance.

	Instruct		Think	
Scale	**Lure %**	**Δ vs. 7B**	**Lure %**	**Δ vs. 7B**
7B	14.9	—	0.9	—
32B	19.6	+4.7pp	0.4	−0.5pp

Table 8: Peak probe AUC across scale×training recipe (OLMo family). The representational gap (Instruct − Think) *shrinks* from 0.034 at 7B to 0.0021 at 32B, even as the behavioral gap widens.

Scale	Instruct AUC	Think AUC	Δ (Inst.−Think)
7B	0.996 (L24)	0.962 (L22)	0.034
32B	0.9999 (L20)	0.9978 (L20)	0.0021

Key finding: asymmetric scale effect. Scale has opposite effects depending on training recipe. For standard (Instruct) models, 4.5× more parameters *increases* vulnerability by 4.7pp (14.9%→19.6%). For reasoning (Think) models, the same scale increase produces no meaningful change (−0.5pp, from 0.9% to 0.4%—a single item difference). This asymmetry confirms that reasoning training provides a qualitatively different form of protection that does not erode with scale.

Diverging behavioral and representational gaps. The behavioral gap between Instruct and Think *widens* with scale (14.0pp at 7B → 19.2pp at 32B), yet the representational gap *shrinks* (0.034 → 0.0021). At 32B, both models encode conflict/control distinctions with near-identical fidelity, but only Think translates this encoding into correct responses. This dissociation reinforces the "readable but not writable" pattern: scale amplifies the encoding of vulnerability-relevant representations in standard models without improving resolution, while reasoning training makes the existing encoding actionable.

C SAE Features and Attention Entropy

This appendix provides full details on the sparse autoencoder (SAE) feature analysis and attention entropy analysis summarized in Section 5.6.

C.1 SAE Reconstruction Fidelity

We apply a Goodfire SAE (65,536 features) to Llama-3.1-8B-Instruct layer 19 residual stream activations. Table 9 reports reconstruction fidelity on our benchmark items.

Table 9: SAE reconstruction fidelity on Llama-3.1-8B-Instruct L19 activations (n = 470 items, `P0` position).

Metric	Value
MSE (reconstruction)	0.011
Explained variance (EV)	74.0%
Mean L0 (active features)	47.9 / 65,536
SAE width	65,536

The explained variance of 74% is moderate: the SAE captures the majority of activation variance but not all. Feature-level findings should be interpreted as a lower bound on the true signal, since reconstruction loss may attenuate feature activations.

C.2 Significant Features

After BH-FDR correction ($q < 0.05$) across all 65,536 features, we identify **41 features** with significantly different mean activation between conflict and control items. These features constitute 0.063% of the total SAE dictionary, indicating high specificity.

C.3 Ma et al. Falsification Protocol

All 41 significant features were subjected to the Ma et al. [Ma et al., 2026] falsification protocol, which tests whether a feature's activation is driven by token-level artifacts rather than context-level semantics.

Protocol. For each feature:

1. Identify the top-3 tokens that maximally activate the feature across all benchmark items.
2. Inject each token into 100 random non-cognitive-bias texts (drawn from a held-out corpus of Wikipedia paragraphs, news articles, and code snippets).
3. Measure the feature's activation in the random context.
4. A feature is declared *spurious* if its mean activation in random contexts exceeds 50% of its mean activation in the benchmark context.

Table 10: SAE falsification results (Llama-3.1-8B-Instruct, L19).

Metric	Value
Features tested (post-FDR)	41
Spurious (failed falsification)	0
Surviving features	**41**
Spurious rate	0.0%

All 41 features survive falsification. The 0% spurious rate is noteworthy given that Méloux et al. [2025] report 45–90% spurious rates for purported "reasoning features" in other settings. The low rate here likely reflects our strict initial filtering (BH-FDR $q < 0.05$ with effect-size thresholds) and the fact that cognitive bias processing engages contextual rather than token-level representations.

Trigger token analysis. Inspection of the top-activating tokens for the 41 surviving features reveals that they are predominantly content words related to probability, quantity, and social categorization (e.g., "likely," "percent," "group," "typical") rather than function words or formatting tokens. This is consistent with the features encoding aspects of probabilistic reasoning and category-based judgment that are engaged specifically by the cognitive bias stimuli.

C.4 Attention Entropy: Full Head-Level Analysis

We compare attention entropy patterns between Llama-3.1-8B-Instruct and R1-Distill-Llama-8B at the `P0` position. Attention entropy is computed as the Shannon entropy of each head's attention distribution, normalized by $\log_2(t)$ where t is the sequence length.

Key findings.

- R1-Distill has nearly **twice the proportion of S2-like-specialized heads** (5.6% vs. 2.9%) compared to Llama. This gap widens further at the KV-group level (9.4% vs. 5.1%), where GQA non-independence is accounted for.
- The majority of significant heads in both models show *control-higher* entropy (82.7% for Llama, 77.7% for R1-Distill), meaning that conflict items generally elicit more focused (lower entropy) attention patterns. This is consistent with conflict items triggering targeted processing of specific diagnostic elements in the prompt.
- Among S2-like-specialized heads, nearly all show conflict-higher entropy (96.7% for Llama, 82.5% for R1-Distill). These heads attend more broadly on conflict items, potentially integrating information across the full problem statement rather than fixating on heuristic-relevant features.

Table 11: Summary of head-level attention entropy tests (32 layers $\times$ 32 heads = 1,024 total per model). Significance assessed via rank-biserial correlation (r_{rb}) between conflict and control items, BH-FDR corrected at $q < 0.05$. S2-like-specialized heads additionally require $|r_{rb}| \geq 0.3$.

Metric	Llama 8B-Inst.	R1-Distill
Total heads tested	1,024	1,024
Significant (BH $q < 0.05$)	473 (46.2%)	515 (50.3%)
Of which conflict-higher	82 (17.3%)	115 (22.3%)
Of which control-higher	391 (82.7%)	400 (77.7%)
S2-like-specialized heads ($q < 0.05$ and $\|r_{rb}\| \geq 0.3$)	30 (2.9%)	57 (5.6%)
Of which conflict-higher	29 (96.7%)	47 (82.5%)
KV-group level	13/256 (5.1%)	24/256 (9.4%)

Layer-wise entropy gap. The entropy gap (conflict $-$ control mean normalized entropy) shows distinct profiles:

- **Llama-3.1-8B:** Peak entropy gap at L7 ($\Delta = -0.016$), concentrated in early-to-mid layers (L4–L14).
- **R1-Distill:** Peak entropy gap at L11 ($\Delta = -0.017$), with a broader distribution extending through L15 and a secondary focus in late layers (L27–L31).

The negative sign indicates that conflict items elicit lower entropy (more focused attention) than control items. The broader distribution in R1-Distill is consistent with reasoning distillation distributing conflict-processing across more layers rather than concentrating it in early attention heads.

D Multi-Seed Robustness

This appendix reports multi-seed evaluation results for Llama-3.1-8B-Instruct and R1-Distill-Llama-8B under sampled decoding (temperature 0.7, top-p = 0.95, 3 seeds: 0, 42, 123). All main-text results use greedy decoding ($T = 0$); this analysis tests whether behavioral findings are robust to the stochasticity introduced by sampling.

D.1 Exact Generation Parameters

Table 12 reports the exact generation keyword arguments used for each model family.

Table 12: Generation keyword arguments for all models. "Greedy" denotes the main-text configuration; "Sampled" denotes the robustness configuration used in this appendix.

Parameter	Greedy (main text)	Sampled (this appendix)
`do_sample`	`False`	`True`
`temperature`	1.0	0.7
`top_p`	—	0.95
`max_new_tokens`	128 (standard) / 2048 (reasoning)	same
System prompt	none	none

Vendor-recommended settings. Our greedy configuration departs from several vendor recommendations: DeepSeek recommends `temperature=0.5-0.7` for R1 models; Qwen recommends `temperature=0.6` with `top_p=0.95` for Qwen 3. We use deterministic greedy decoding throughout the main text to (a) ensure exact reproducibility without seed dependence, (b) isolate representational differences from generation-strategy effects, and (c) simplify the probing analysis, since `P0` activations are deterministic given the prompt regardless of downstream sampling. This appendix demonstrates that the qualitative findings— in particular the S1/S2 lure-rate gap and the stability of probe AUC at `P0`—are preserved under vendor-recommended sampled decoding (`temperature=0.7`).

Robustness argument. The key robustness claim is twofold: (1) probe results at P0 are *invariant* to generation strategy because P0 activations depend only on the prompt, not on downstream sampling; and (2) behavioral lure-rate gaps between S1 and S2 models are qualitatively preserved under sampled decoding, though the magnitudes shift for reasoning models whose chain-of-thought is sensitive to stochastic token selection (Section D.4).

D.2 Overall Stability

Table 13: Multi-seed overall lure rates (11 categories, 235 conflict items per seed, sampled decoding at $T = 0.7$). Main-text greedy results are shown for comparison. "σ" = standard deviation across seeds in percentage points.

	Sampled ($T = 0.7$, **3 seeds**)				**Greedy**
Model	**Seed 0**	**Seed 42**	**Seed 123**	**Mean** $\pm\ \sigma$	($T = 0$)
Llama-3.1-8B-Inst.	26.4%	27.2%	28.9%	27.5% $\pm$ 1.3pp	27.3%
R1-Distill-Llama-8B	11.5%	11.5%	13.2%	12.1% $\pm$ 1.0pp	2.4%*

*The large gap between R1-Distill greedy (2.4%) and sampled (12.1%) reflects the generation strategy dependence discussed in Section D.4.

Llama stability. Llama's overall lure rate is remarkably stable across seeds: 27.5% $\pm$ 1.3pp, closely matching the greedy result of 27.3%. This confirms that the behavioral vulnerability is a robust property of the model's representations, not an artifact of a particular decoding path.

R1-Distill: greedy vs. sampled divergence. R1-Distill shows a large gap between greedy (2.4%) and sampled (12.1%) lure rates. Under sampled decoding, the reasoning model's chain-of-thought occasionally takes suboptimal paths that fail to override heuristic lures. The within-seed stability is high (11.5%–13.2%, $\sigma = 1.0$pp), confirming this is a systematic effect of the generation strategy rather than random noise.

D.3 Per-Category Profiles Under Sampling

Table 14 reports per-category lure rates under sampled decoding.

Table 14: Per-category lure rates (%) under sampled decoding ($T = 0.7$, mean $\pm\ \sigma$ across 3 seeds). "Δ vs. greedy" = sampled mean $-$ greedy. Categories marked with † show instability ($\sigma > 10$pp).

	Llama		**R1-Distill**	
Category	**Sampled**	Δ **greedy**	**Sampled**	Δ **greedy**
Base rate	81.0 $\pm$ 3.3	-3.0	1.0 $\pm$ 1.6	-3.0
Conjunction	3.3 $\pm$ 2.9	-51.7	11.7 $\pm$ 2.9	$+11.7$
Syllogism	41.3 $\pm$ 9.2	-10.7	14.7 $\pm$ 8.3	$+14.7$
CRT variants	35.6 $\pm$ 5.1	$+35.6$	35.6 $\pm$ 5.1	$+32.6$
Arithmetic	4.0 $\pm$ 0.0	$+4.0$	5.3 $\pm$ 2.3	$+5.3$
Framing	53.3 $\pm$ 2.9	$+53.3$	31.7 $\pm$ 17.6†	$+31.7$
Anchoring	0.0 $\pm$ 0.0	0.0	1.7 $\pm$ 2.9	-8.3
Sunk cost	4.4 $\pm$ 3.8	$+4.4$	0.0 $\pm$ 0.0	0.0
Loss aversion	0.0 $\pm$ 0.0	0.0	22.2 $\pm$ 10.2†	$+22.2$
Certainty eff.	6.7 $\pm$ 6.7	$+6.7$	0.0 $\pm$ 0.0	0.0
Availability	8.9 $\pm$ 10.2†	-4.1	0.0 $\pm$ 0.0	0.0

Category profile shifts. While overall lure rates are stable, category profiles shift substantially under sampled decoding. For Llama, the most dramatic changes are:

- **CRT and framing become highly vulnerable** (0% greedy $\rightarrow$ 35.6% and 53.3% sampled respectively). These categories are immune under greedy decoding because the maximum-probability token path avoids the lure, but sampling occasionally selects lower-probability heuristic responses.

- **Conjunction drops dramatically** (55% greedy $\rightarrow$ 3.3% sampled), suggesting that Llama's greedy conjunction errors are a narrow peak in the output distribution that sampling diversifies away from.

For R1-Distill, sampling reveals latent vulnerabilities that greedy decoding masks: CRT (3% $\rightarrow$ 35.6%), framing (0% $\rightarrow$ 31.7%), and loss aversion (0% $\rightarrow$ 22.2%) all emerge as substantially vulnerable when the chain-of-thought can take suboptimal reasoning paths.

D.4 Generation Strategy Dependence

Table 15: Generation strategy comparison. Greedy results represent the maximum-probability decoding path; sampled results represent the distributional expectation.

Model	Greedy	Sampled	$\lvert\Delta\rvert$
Llama-3.1-8B-Inst.	27.3%	27.5% $\pm$ 1.3pp	0.2pp
R1-Distill-Llama-8B	2.4%	12.1% $\pm$ 1.0pp	9.7pp

Implications. The greedy–sampled divergence is asymmetric: Llama's vulnerability is essentially identical under both strategies ($|\Delta| = 0.2$pp), while R1-Distill's vulnerability is 4$\times$ higher under sampling ($|\Delta| = 9.7$pp). This asymmetry has two implications:

1. **Greedy results are maximum-probability, not distributional.** The 2.4% greedy lure rate for R1-Distill reflects the single most-likely decoding path. The 12.1% sampled rate better approximates the model's distributional vulnerability—the expected lure rate over the space of possible reasoning traces.
2. **Reasoning models are more sensitive to generation strategy.** The reasoning model's chain-of-thought introduces a long sequence of stochastic choices, each of which can derail the reasoning process. Standard models generate shorter responses with fewer opportunities for the decoding strategy to influence the outcome.

D.5 Probe Invariance to Generation Strategy

Crucially, probe results at `P0` are **unaffected** by generation strategy, because `P0` activations are deterministic given the prompt. The pre-decision representation is identical regardless of whether the model subsequently generates via greedy or sampled decoding. This confirms that our probing results (AUC values, cross-prediction, steering) reflect stable properties of the model's internal representations rather than artifacts of a particular generation strategy.

The probe–behavior divergence under sampling further supports the "conflict detection without resolution" interpretation: the model *always* encodes the conflict in its `P0` representation (AUC = 0.930 regardless of decoding), but whether it successfully *resolves* that conflict depends on the stochastic reasoning path it takes during generation.

E Token-Length Confound Analysis

Near-perfect probe AUC (e.g., OLMo-32B at 0.9999) can in principle arise from trivial confounds like systematic prompt-length differences between conflict and control items. We audit length systematically to rule this out.

Per-category length statistics. Table 16 reports the mean Llama tokenizer subword count per (category, condition), Cohen's d for conflict vs. control, and the Welch t-test p-value. Length differences are substantial on base rate ($d = +3.69$) and conjunction ($d = +6.36$), balanced on syllogism ($d = +0.02$), and reversed on framing ($d = -4.86$) and certainty effect ($d = -4.64$).

Why length is not the primary signal. Three observations rule out length as the dominant confound. First, a probe trained on syllogism alone (where length is balanced, $d = +0.02$) achieves within-category AUC = 0.936 and transfers to base rate (0.950) and conjunction (0.990)—the shared direction this probe recovers cannot be length. Second, length direction is inconsistent across

Table 16: Prompt subword length (Llama-3.1-8B tokenizer) for conflict vs. control items by category. "d" is Cohen's d; positive values indicate conflict items are longer; p is from Welch's t-test.

Category	Conflict mean	Control mean	Cohen's d	p
Vulnerable (probe training):				
Base rate	98.0	66.7	$+3.69$	$< 10^{-4}$
Conjunction	87.7	63.4	$+6.36$	$< 10^{-4}$
Syllogism	93.3	93.2	$+0.02$	0.953
Original immune (cross-prediction):				
CRT	47.0	51.8	-0.44	0.097
Arithmetic	70.9	51.0	$+6.21$	$< 10^{-4}$
Framing	97.2	106.2	-4.86	$< 10^{-4}$
Anchoring	60.4	45.5	$+4.62$	$< 10^{-4}$
Added categories:				
Availability	41.0	46.5	-2.26	$< 10^{-4}$
Certainty effect	56.1	65.3	-4.64	$< 10^{-4}$
Loss aversion	73.3	58.6	$+5.23$	$< 10^{-4}$
Sunk cost	176.6	114.2	$+4.86$	$< 10^{-4}$
Overall (all 470)	**81.1**	**68.4**	$+0.45$	$< 10^{-4}$

categories: conjunction/base rate have $d > 0$ while framing/certainty effect have $d < 0$, so a pure-length probe would produce maximally anti-correlated cross-category predictions, not the "immune-cluster compression" pattern we observe in the logit-histogram analysis (Section 5.2). Third, the sentence-transformer text baseline, which has direct access to length as a feature, caps at 0.840; the 16pp gap to the activation probe (0.999) is information not extractable from the prompt text alone.

Specific OLMo-32B audit. OLMo-3.1-32B-Instruct achieves probe AUC = 0.9999 on the vulnerable three categories. Since syllogism (length-balanced) is one of those three, the probe cannot be purely length-driven: if it were, it would collapse to chance on syllogism and drag the joint AUC below 0.67. The high joint AUC implies the probe separates syllogism conflict vs. control at high accuracy despite near-identical prompt lengths (93.3 vs. 93.2 tokens).

F Full Behavioral Breakdown (All Models, All Categories)

Table 17 provides the complete behavioral breakdown that the main-text Table 1 summarizes. For each of the eight evaluated models and each of the eleven categories, we report: conflict accuracy (Acc), S1-lure rate (Lure), other-response rate (Other), matched-control accuracy (Ctrl), and accuracy gap ($=$ Ctrl $-$ Acc). Brackets give Wilson 95% confidence intervals. The final column reports an overall McNemar mid-p test of conflict vs. matched-control accuracy, which uses the paired structure of the benchmark directly.

This table addresses reviewer requests to (i) report the full response distribution rather than just lure rate, (ii) include matched-control accuracy and conflict–control gap for every model, and (iii) use the matched-pair structure statistically. Dashes indicate categories not evaluated on a particular model (e.g., Qwen and R1-Distill were run on the original 330-item, 7-category subset; OLMo models run on the full 470 items).

G Steering Specificity Controls

Table 18 reports the full sweep of the specificity controls discussed in Section 5.3 for Llama-3.1-8B-Instruct at layer 14. Four sweeps were run: (1) probe-direction steering on the 80 vulnerable-category conflict items (replication of the main-text result); (2) probe-direction steering on the 80 matched non-conflict items (specificity test); (3) mean-difference direction $\bar{\mathbf{h}}_{\text{conflict}} - \bar{\mathbf{h}}_{\text{control}}$ steering on the conflict items (CogBias baseline); (4) mean-difference direction steering on the control items. All use continuous steering (hook active across the whole 128-token generation). The cosine similarity between the probe-weight vector and the mean-difference vector is only 0.314, meaning they are

Table 17: Complete behavioral results across models and categories. Conflict accuracy (Acc), lure rate (Lure), and other-response rate (Other) on conflict items; control accuracy (Ctrl); and accuracy gap (Gap = Ctrl − Acc). Wilson 95% CIs in brackets. McNemar mid-p tests conflict vs. matched control accuracy.

Model	Metric	CRT	Base Rate	Conjunction	Syllogism	Arithmetic	Framing	Anchoring	Sunk Cost	Av
Llama-3.1-8B	Acc	93.3% [79,98]	16.0% [6,35]	45.0% [26,66]	48.0% [30,67]	100.0% [87,100]	100.0% [84,100]	50.0% [30,70]	—	
	Lure	0.0% [0,11]	**84.0% [65,94]**	**55.0% [34,74]**	**52.0% [33,70]**	0.0% [0,13]	0.0% [0,16]	0.0% [0,16]	—	
	Other	6.7%	0.0%	0.0%	0.0%	0.0%	0.0%	50.0%	—	
	Ctrl	90.0% [74,97]	100.0% [87,100]	100.0% [84,100]	52.0% [33,70]	100.0% [87,100]	100.0% [84,100]	80.0% [58,92]	—	
	Gap	-3.3pp	+84.0pp	+55.0pp	+4.0pp	+0.0pp	+0.0pp	+30.0pp	—	
R1-Distill-8B	Acc	83.3% [66,93]	16.0% [6,35]	100.0% [84,100]	100.0% [87,100]	96.0% [80,99]	100.0% [84,100]	65.0% [43,82]	—	
	Lure	3.3% [1,17]	4.0% [1,20]	0.0% [0,16]	0.0% [0,13]	0.0% [0,13]	0.0% [0,16]	10.0% [3,30]	—	
	Other	13.3%	80.0%	0.0%	0.0%	4.0%	0.0%	25.0%	—	
	Ctrl	100.0% [89,100]	16.0% [6,35]	100.0% [84,100]	96.0% [80,99]	100.0% [87,100]	100.0% [84,100]	65.0% [43,82]	—	
	Gap	+16.7pp	+0.0pp	+0.0pp	-4.0pp	+4.0pp	+0.0pp	+0.0pp	—	
Qwen3-8B (no think)	Acc	73.3% [56,86]	44.0% [27,63]	5.0% [1,24]	48.0% [30,67]	100.0% [87,100]	100.0% [84,100]	80.0% [58,92]	—	
	Lure	0.0% [0,11]	**56.0% [37,73]**	**95.0% [76,99]**	0.0% [0,13]	0.0% [0,13]	0.0% [0,16]	10.0% [3,30]	—	
	Other	26.7%	0.0%	0.0%	52.0%	0.0%	0.0%	10.0%	—	
	Ctrl	100.0% [89,100]	100.0% [87,100]	100.0% [84,100]	52.0% [33,70]	100.0% [87,100]	100.0% [84,100]	65.0% [43,82]	—	
	Gap	+26.7pp	+56.0pp	+95.0pp	+4.0pp	+0.0pp	+0.0pp	-15.0pp	—	
Qwen3-8B (think)	Acc	76.7% [59,88]	96.0% [80,99]	45.0% [26,66]	100.0% [87,100]	96.0% [80,99]	100.0% [84,100]	75.0% [53,89]	—	
	Lure	0.0% [0,11]	4.0% [1,20]	**55.0% [34,74]**	0.0% [0,13]	0.0% [0,13]	0.0% [0,16]	0.0% [0,16]	—	
	Other	23.3%	0.0%	0.0%	0.0%	4.0%	0.0%	25.0%	—	
	Ctrl	93.3% [79,98]	100.0% [87,100]	100.0% [84,100]	100.0% [87,100]	100.0% [87,100]	100.0% [84,100]	90.0% [70,97]	—	
	Gap	+16.7pp	+4.0pp	+55.0pp	+0.0pp	+4.0pp	+0.0pp	+15.0pp	—	
OLMo-3-7B-Instruct	Acc	73.3% [56,86]	54.3% [38,70]	50.0% [30,70]	100.0% [87,100]	80.0% [61,91]	100.0% [84,100]	35.0% [18,57]	100.0% [80,100]	86.7
	Lure	3.3% [1,17]	**45.7% [30,62]**	**50.0% [30,70]**	0.0% [0,13]	0.0% [0,13]	0.0% [0,16]	5.0% [1,24]	0.0% [0,20]	**13.**
	Other	23.3%	0.0%	0.0%	0.0%	20.0%	0.0%	60.0%	0.0%	
	Ctrl	90.0% [74,97]	100.0% [90,100]	100.0% [84,100]	100.0% [87,100]	96.0% [80,99]	100.0% [84,100]	50.0% [30,70]	0.0% [0,20]	40.0
	Gap	+16.7pp	+45.7pp	+50.0pp	+0.0pp	+16.0pp	+0.0pp	+15.0pp	-100.0pp	
OLMo-3-7B-Think	Acc	96.7% [83,99]	100.0% [90,100]	100.0% [84,100]	72.0% [52,86]	88.0% [70,96]	100.0% [84,100]	80.0% [58,92]	46.7% [25,70]	100.0
	Lure	3.3% [1,17]	0.0% [0,10]	0.0% [0,16]	4.0% [1,20]	0.0% [0,13]	0.0% [0,16]	0.0% [0,16]	0.0% [0,20]	0
	Other	0.0%	0.0%	0.0%	24.0%	12.0%	0.0%	20.0%	53.3%	
	Ctrl	93.3% [79,98]	100.0% [90,100]	100.0% [84,100]	68.0% [48,83]	100.0% [87,100]	100.0% [84,100]	70.0% [48,85]	0.0% [0,20]	100.0
	Gap	-3.3pp	+0.0pp	+0.0pp	-4.0pp	+12.0pp	+0.0pp	-10.0pp	-46.7pp	
OLMo-3.1-32B-Instruct	Acc	86.7% [70,95]	25.7% [14,42]	50.0% [30,70]	96.0% [80,99]	96.0% [80,99]	70.0% [48,85]	70.0% [48,85]	100.0% [80,100]	100.0
	Lure	6.7% [2,21]	**74.3% [58,86]**	**50.0% [30,70]**	0.0% [0,13]	0.0% [0,13]	**30.0% [15,52]**	10.0% [3,30]	0.0% [0,20]	0
	Other	6.7%	0.0%	0.0%	4.0%	4.0%	0.0%	20.0%	0.0%	
	Ctrl	93.3% [79,98]	100.0% [90,100]	95.0% [76,99]	88.0% [70,96]	72.0% [52,86]	90.0% [70,97]	85.0% [64,95]	0.0% [0,20]	33.3
	Gap	+6.7pp	+74.3pp	+45.0pp	-8.0pp	-24.0pp	+20.0pp	+15.0pp	-100.0pp	
OLMo-3.1-32B-Think	Acc	100.0% [89,100]	100.0% [90,100]	100.0% [84,100]	100.0% [87,100]	100.0% [87,100]	100.0% [84,100]	80.0% [58,92]	80.0% [55,93]	100.0
	Lure	0.0% [0,11]	0.0% [0,10]	0.0% [0,16]	0.0% [0,13]	0.0% [0,13]	0.0% [0,16]	0.0% [0,16]	6.7% [1,30]	0
	Other	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	20.0%	13.3%	
	Ctrl	100.0% [89,100]	100.0% [90,100]	100.0% [84,100]	100.0% [87,100]	100.0% [87,100]	100.0% [84,100]	85.0% [64,95]	0.0% [0,20]	100.0
	Gap	+0.0pp	+0.0pp	+0.0pp	+0.0pp	+0.0pp	+0.0pp	+5.0pp	-80.0pp	

substantially different directions in the 4096-dimensional residual stream despite both encoding conflict-vs.-control information to some degree.

Three observations. (i) The probe direction produces *zero* lure responses on control items at any α: the direction is specific to the S1/S2 conflict subspace and does not generically perturb answers. The small drop in correctness at $\alpha = -5$ on control items (78.75%) reflects residual-stream coherence loss (the 21.25% of items become "other" responses, not lures), the same pattern seen with random directions in the main text. (ii) The probe-weight direction is *bidirectional*: moving toward S1 (negative α) increases lure by 16.25pp, moving toward S2 (positive α) decreases lure by 21.25pp. Bidirectional dose-response is stronger evidence of causal sufficiency than a unidirectional effect. (iii) The mean-difference baseline is *unidirectional*: negative α has essentially no effect (lure stays around 53–55%), while positive α strongly reduces lure (to 11.25% at $\alpha = +5$). Mean-difference steering efficiently improves behavior but cannot test causal sufficiency by worsening it—a key methodological motivation for probe-weight steering.

Table 18: Steering specificity controls on Llama-3.1-8B-Instruct, L14. "Probe" is the logistic-regression weight vector $\mathbf{w}$; "Mean-diff" is $\bar{\mathbf{h}}_{\text{conflict}} - \bar{\mathbf{h}}_{\text{control}}$ (CogBias-style). "Conflict" = 80 vulnerable-category conflict items (each has an S1 lure answer); "Control" = 80 matched non-conflict items (each has a single normative answer).

Direction	**Items**	$\alpha = -5$	-3	-1	0	$+1$	$+3$	$+5$
Lure rate (%)								
Probe	Conflict	68.75	61.25	56.25	52.50	48.75	41.25	31.25
Probe	Control	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Mean-diff	Conflict	53.75	55.00	55.00	52.50	36.25	21.25	11.25
Mean-diff	Control	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Correct rate (%)								
Probe	Conflict	31.25	38.75	43.75	47.50	51.25	58.75	68.75
Probe	Control	78.75	87.50	98.75	100.00	100.00	100.00	100.00
Mean-diff	Conflict	46.25	45.00	45.00	47.50	63.75	78.75	88.75
Mean-diff	Control	60.00	73.75	97.50	100.00	100.00	100.00	100.00